

README

DocProcessor

Documentation processing and feature map extraction for QA automation.

Overview

DocProcessor is a standalone Go module that loads project documentation, builds structured feature maps, and tracks verification coverage. It is designed to work with LLM agents for intelligent feature extraction, but also includes heuristic-based extraction for offline use.

Quick Start

```
# Clone
git clone git@github.com:vasic-digital/DocProcessor.git
cd DocProcessor
```

```
# Build
go build ./...
```

```
# Test
go test ./... -race -count=1
```

```
# Run
go run ./cmd/docprocessor /path/to/docs
```

Architecture

DocProcessor is organized into 6 packages:

Package	Purpose
pkg/loader	Document loading and parsing (Markdown, YAML, HTML, AsciiDoc, RST)

Package	Purpose
pkg/ feature	Feature extraction, FeatureMap building, FeatureMapBuilder
pkg/ coverage	Thread-safe coverage tracking with RWMutex
pkg/ docgraph	Inter-document link graph with JSON/Mermaid export
pkg/llm	LLMAgent interface and prompt templates
pkg/config	Configuration from .env files

Processing Pipeline

Load Docs -> Parse Sections -> Extract Features -> Build FeatureMap -> Enrich (LLM) -> Track Coverage

1. **Load & Parse** -- Scan project tree for documentation files
2. **Extract Features** -- Heuristic extraction or LLM-powered extraction
3. **Build Feature Map** -- Structured, queryable map with categories and platform matrix
4. **Enrich** -- LLM infers screens and generates test steps
5. **Track Coverage** -- Thread-safe per-platform verification tracking

Key Interfaces

- loader.Loader -- Load documents from filesystem
- feature.FeatureMapBuilder -- Build feature maps from documents
- coverage.CoverageTracker -- Track feature verification status
- llm.LLMAgent -- Injected LLM for intelligent extraction (no hard dependency)

Configuration

Copy .env.example to .env and customize:

```
HELIX_DOCS_ROOT=./docs
HELIX_DOCS_AUTO_DISCOVER=true
HELIX_DOCS_FORMATS=md,yaml,html,adoc,rst
```

Testing

```
make test          # Run tests
make test-race     # Run with race detection
make test-cover    # Run with coverage report
```

190+ tests across 6 test types: unit, integration, stress, security, E2E, automation.

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